

resumes. The disadvantage is most severe in business-related fields such as accounting, sales, and finance, and less pronounced in areas like agriculture, arts, and automotive. If similar advantages were to persist across repeated hiring cycles, they could give rise to a “lock-in” effect, in which the stylistic patterns favored by dominant LLMs gradually become entrenched in applicant pools, with the potential to amplify inequities and reduce diversity in candidate selection.

To mitigate AI self-preference, we propose two simple yet effective strategies that directly target the underlying mechanism of self-recognition—a model’s ability to implicitly identify content it generated. The first strategy uses system prompting to explicitly instruct models to ignore the origin of resumes and focus only on substantive content. The second strategy employs a majority voting ensemble, combining the evaluator model with smaller models that exhibit weaker self-recognition, thereby diluting the bias of any single LLM. Across all tested LLMs, these interventions reduce LLM-versus-human self-preference by 17% to 63% in relative terms. These results demonstrate that while self-preference bias is widespread and consequential, it is not immutable: straightforward design interventions can substantially improve fairness in LLM-based hiring operations.

Together, these findings advance our understanding of how generative AI systems shape operational decision processes in organizations. We document and quantify self-preference bias in the context of resume screening by developing a measurement framework grounded in established fairness metrics. In addition, we show that self-preference bias can be substantially reduced through targeted interventions informed by LLM’s self-recognition behavior. In doing so, the study introduces a novel and practical perspective on AI fairness for business operations—one that moves beyond concerns about demographic disparities to address interactional biases that arise when AI systems evaluate content they themselves could have produced.

The remainder of the paper is organized as follows. We begin by reviewing the related literature in Section 2. Section 3 defines and outlines the measurement of AI self-preferencing bias. Section 4 describes the dataset and experimental design, and Section 5 presents the empirical findings. In Section 6, we introduce and evaluate mitigation strategies. Finally, we conclude in Section 7 with a discussion of key implications and directions for future research.

2. Literature Review

Our research contributes to multiple strands of literature: (1) screening and hiring as operational processes, (2) generative AI in labor market and workplace operations, (3) algorithmic fairness, bias, and responsible operations, and (4) LLMs as judges in decision making.

2.1. Screening and Hiring as Operational Processes

Operations Management (OM) research has long examined hiring and screening as core operational processes through which organizations allocate scarce evaluation capacity and form productive

matches under uncertainty (e.g., Freeman 1983, Jovanovic 1979, Arlotto et al. 2014, Yoo et al. 2016, Koren 2026). This work positions workforce selection at the interface between operations and human resource management (Boudreau et al. 2003), conceptualizing recruitment and screening not as purely administrative functions but as operational processes through which firms transform applicant pools into productive labor inputs that support day-to-day operations.

A central implication of this lens is that hiring pipelines are typically *multi-stage* and *capacity constrained*. Upstream screening (e.g., resume review) funnels a large applicant pool into a limited set of downstream evaluations (e.g., interviews and assessments), where time, managerial attention, and specialist resources are scarce. As in other service and information-processing systems, errors at upstream triage have asymmetric and persistent consequences: false negatives irreversibly eliminate qualified candidates, while false positives consume scarce evaluation capacity and degrade downstream match quality. Recent work in platform and labor market operations further emphasizes how design choices in these screening stages shape congestion, equilibrium outcomes, and match efficiency. For example, Farajollahzadeh et al. (2025) show that interview-stage diversity interventions such as the Rooney Rule can backfire in capacity-constrained settings with noisy pre-interview signals, while Baek et al. (2025) demonstrate that strategic interview decisions in multistage, multi-firm hiring processes can substantially mitigate market inefficiencies and increase the number of hires overall.

Our paper contributes to this stream of literature by identifying a new and operationally consequential distortion that arises at the *earliest* stage of the hiring funnel when screening is delegated to AI evaluators. Existing OM models implicitly assume that evaluators process signals in a manner that is neutral to how those signals are generated. We relax this assumption and offer empirical evidence by studying an environment in which applicants increasingly use LLMs to generate screening inputs (e.g., resumes), while firms deploy similar models to evaluate those same inputs. We show that AI evaluators can exhibit systematic self-preference, favoring content generated by the same model, even when underlying candidate information is held constant. This interaction-driven bias is distinct from standard screening noise: it is endogenous to pipeline design choices (which tools are used upstream and downstream) and can create path-dependent lock-in of “dominant” resume styles, with implications for both operational performance (misranking and misallocation) and responsible hiring system design.

2.2. Generative AI in Labor Market and Workplace Operations

A growing literature examines the role of AI in labor market and workplace operations. Research spanning operations management, economics, and information systems documents how AI tools are increasingly deployed in both production and evaluation tasks, with important implications for

worker performance, learning, and market outcomes. For example, Brynjolfsson et al. (2025) provide field evidence that a generative AI assistant improves productivity in customer support, with heterogeneous effects depending on workers’ experience and baseline skill. In labor-market contexts, a complementary stream studies how AI-assisted writing affects the formation of applicant signals and subsequent hiring outcomes. Wiles et al. (2025) show that algorithmic resume writing assistance improves writing quality and increases hiring, suggesting that generative AI can meaningfully alter the informational content and effectiveness of applicant signals in matching markets. At the same time, the introduction of algorithms into evaluation processes raises well-documented concerns. Prior work shows that human decision makers may discount or resist algorithmic recommendations (e.g., Dietvorst et al. 2015), and that algorithmic hiring systems can encode or amplify existing biases when trained on historical data or used with discretion (e.g., Hoffman et al. 2018, Cowgill 2018).

Our paper contributes to this emerging body of work by identifying a distinct and previously unexamined mechanism that arises when generative AI is used on both sides of the hiring market. We provide controlled, large-scale evidence that such coupling can create systematic advantages for certain applicants even when quality is held constant. Specifically, we document and quantify AI self-preferencing in resume screening. We then connect this bias to labor-market consequences by simulating capacity-constrained shortlisting pipelines across occupations, showing how evaluator self-preference affects the allocation of interview opportunities. Finally, we demonstrate that simple interventions (e.g., system prompting and majority-vote ensemble) can substantially mitigate this distortion, offering practical guidance for firms adopting generative AI in hiring workflows.

2.3. Algorithmic Fairness, Bias, and Responsible Operations

Our work also contributes to the growing literature on responsible practices in operations management, which emphasizes that operational decisions should be evaluated not only by efficiency but also by their broader implications for fairness and stakeholder welfare. Recent research highlights that dimensions such as fairness, equity, and employee well-being are central—rather than peripheral—to the design of effective and sustainable operational systems (e.g., Lee and Tang 2018, Netessine 2022, Kesavan et al. 2022, Wang et al. 2023). This body of work underscores a core insight: responsible operations require careful attention to how system design choices shape downstream human outcomes, often in context-specific ways.

Hiring and screening provide a particularly salient application of these ideas. Discrimination in labor markets has long been a central concern for both policymakers and economists, with extensive research documenting that certain race, gender, and age subpopulations face unfair treatment from recruiters in the hiring process (e.g., Bertrand and Mullainathan 2004, Kline et al. 2022, Neumark